

3-SPR-storage-nodes-6-IceLake-clients

Submitter: **Lightbits Labs**
Solution type: **Software-Defined NVMe/TCP Block Storage**

Submission date: **July-10-2025**

Benchmark version: **mlpstorage 2.0.0b1**
Submission division: **Closed**

Hardware

6x Client servers

- 2x Intel® Xeon® Gold 6338 CPU
- 512GB RAM
- 2x100Gbps NICs (Intel e810)

3x Storage servers

- 2x Intel® Xeon® Platinum 8480+ CPU
- 512GB RAM
- 2x100Gbps NICs (Mellonox Connect-X 5)
- 16x7.68TB Micron 7500 pro NVMe SSD
- 2 Lightbits instances per server
- Dual NUMA per server

Storage System - Power & Rack Units

Component	Power Supply Unit	Nameplate Power	Design Power	Rack Units
Storage Server 1	PS 1	2600 W	1080 W	2
Storage Server 2	PS 1	2600 W	1080 W	2
Storage Server 3	PS 1	2600 W	1080 W	2
Total		**7800 W**	**3240 W**	**6**

Software

Client servers

- Ubuntu 22.04.4 LTS
- 5.15.0-143-generic
- nvme-cli 1.16
- Lightbits discovery-client

Storage servers

- Alma Linux 9.4
- 5.14.0-427.13.1.el9_4.x86_64
- Lightbits release 3.14.1

Storage Configuration

Client servers

- Striped LVM with 6 Lightbits volumes
- XFS filesystem on LVM

Storage servers

- 16x NVMe SSD per server (8x SSD per Lightbits instance)
- Compression disabled
- Replication factor: 1
- 6 volumes created (1 in each Lightbits instance) and are available to all clients
- Do not allow cross numa

Networking Configuration

Client servers

- 2x 100gbps NICs (Intel e810)
  - LACP bonding - mode 802.3ad

Storage servers

- 2x 100gbps NICs (Mellonox Connect-X 5)
- 2 Lightbits instances per server
  - 1x 100Gbps NIC per Lightbits instance

There’s no need for a dedicated or special network equipment, just allow data traffic between all storage servers and clients

Network Diagram

